

TERMINOLOGY of the SIMPLEX METHOD

CORNER POINT FEASIBLE SOL: solution is the corners of the
(CPF)
feasible region (i.e., extreme point
or vertex)

REMARK: if a problem have EXACTLY ONE OPTIMAL SOL,
that is a CPF SOL

if it has MULTIPLE SOL, at least two are CPF

ADJACENT CPF SOL: two CPF are ADJACENT if they share
 $n-1$ constraints boundaries, i.e., they are
connected by a line segment that lies in
the constraints boundaries

SLACK VARIABLE: variable that we add to the system of INEQUALITIES
to make it a system of EQUATIONS

AUGMENTED SOLUTION: solution for the original values that has
been ~~exp~~ augmented by the corresponding
values of the slack variable

BASIC SOLUTION: augmented corner point solution

BASIC FEASIBLE SOL: augmented corner point feasible solution

NON-BASIC VARIABLE: variables of the augmented system that we
choose to force to ZERO

BASIC VARIABLE: variables that correspond to the BASIS, hence,
in general, different from zero (not necessarily though)

BASIS: set of basic variables (same as in linear algebra)

2019年12月14日 星期六 晴

BASIC SOLUTIONS and so on...

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix}$$

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5$

• the augmented matrix has full rank (3)
because of the 0-1 pattern created by
the slack variables

=> there exists a subset of 3 columns
which are linearly independent
(col. rank = row. rank)

REMARK: in general this subsets are in a many-to-1

correspondence with corner-point sol (not necessarily feasible)

=> each subset induces a unique corner-point (but a
corner-point might be induced by many subsets)

— in this example is actually 1-to-1 —

EXAMPLE: $\{x_2, x_3, x_5\}$ selected columns

=> $x_1 = 0 \quad x_4 = 0$ and solve for x_2, x_3, x_5

(the solution will be UNIQUE)

$$\begin{cases} x_1 + x_3 = 4 \\ 2x_2 + x_4 = 12 \\ 3x_1 + 2x_3 + x_5 = 18 \end{cases}$$

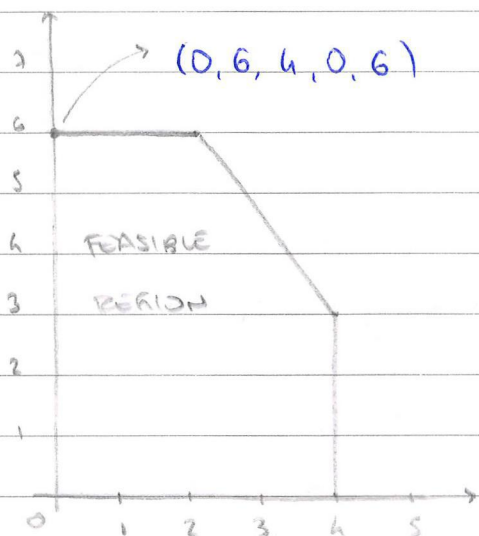
$$\begin{cases} x_3 = 4 \\ 2x_2 = 12 \\ 2x_2 + x_5 = 18 \end{cases} \quad \begin{cases} x_3 = 4 \\ x_2 = 6 \\ x_5 = 18 - 12 = 6 \end{cases}$$

$$(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$$

we can also use the matrix formulation

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 2 & 0 & 0 \\ 2 & 0 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_3 = 4 \\ x_2 = 6 \\ x_5 = 6 \end{cases}$$



$(0, 6, 4, 0, 6)$

solution with values of slack variables x_3, x_4, x_5

- BASIC SOLUTION = exponential corner.
joint solution

- BASIC FEASIBLE SOLUTION because it is feasible for the LP

- BASIS = $\{x_2, x_3, x_5\}$

- BASIC VARIABLES = x_2, x_3, x_5

- NON-BASIC VARIABLES = x_1, x_4

REMARK: • every linear independent subset of 3 columns is a basis and generates a basic solution, although not all will be feasible

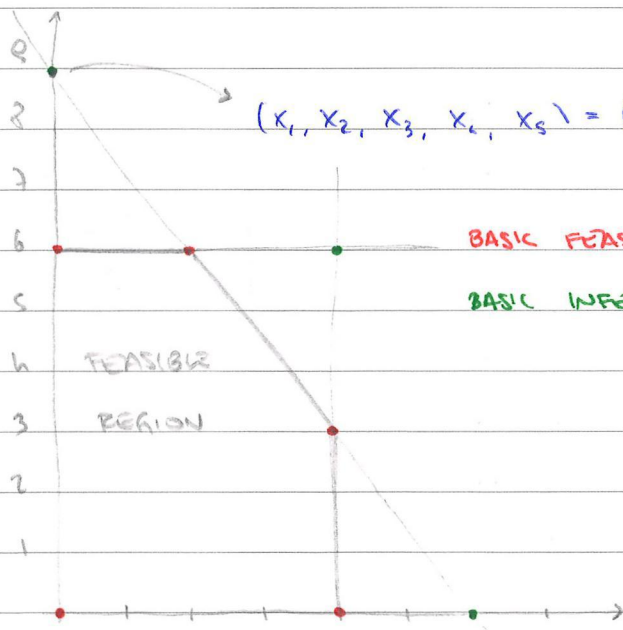
- A basic solution is feasible iff all the decision variables and slack variables are NON-NEGATIVE

EXAMPLE: $\{x_2, x_3, x_4\}$ selected columns

$$x_1 = 0 \quad x_5 = 0$$

$$\begin{cases} x_1 + x_3 = 4 & x_3 = 4 \\ 2x_2 + x_4 = 12 & x_2 = 6 - x_4/2 \\ 3x_1 + 2x_2 + x_5 = 12 & 12 - x_4 = 12 \end{cases}$$

$$\begin{cases} x_4 = -6 ! \\ x_3 = 4 \\ x_2 = 9 \end{cases}$$



$(x_1, x_2, x_3, x_4, x_5) = (0, 9, 4, -6, 0)$

BASIC INFEASIBLE SOL

BASIC FEASIBLE SOLUTIONS

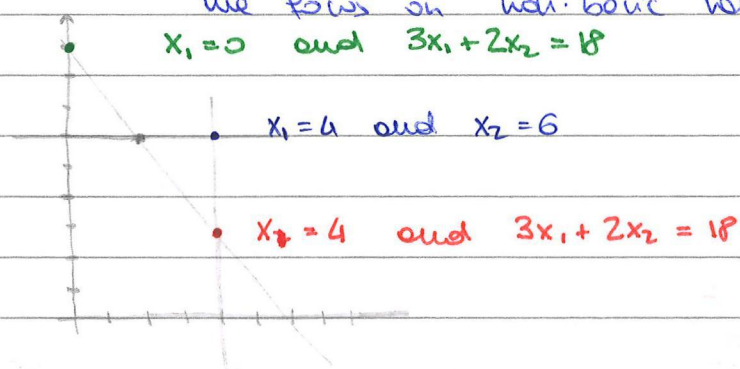
BASIC INFEASIBLE SOLUTIONS

RECAP: • n decision variables
 m constraints (+ non-negativity constraints)
 $\Rightarrow$ AUGMENTED FORM: m equalities
 $n+m$ variables

UNDER DETERMINED SYSTEM

- to obtain a solution we need to **FORCE** n variables to **ZERO** (so that the remaining columns are lin. indep.)
- **GEOMETRIC INTERPRETATION**: we choose n constraints boundaries that should be simultaneously satisfied with equality

Same as the **ALGEBRAIC INTERPRETATION** which focus on basic variables (and basis) while here we focus on non-basic variables



NON-BASIC VARIABLES: variables we force to be zero

$\downarrow$
ZERO by
 definition!

connected 1-to-1 with the subset of functional constraints that we want to be met with equality (**TIGHT**) in order to define a CFF sol

BASIC VARIABLES: those that are NON-BASIC

$\Rightarrow$ they form a **BASIS**

We choose n variables to make non-basic, The basic variables (i.e., the basis) is the set of m remaining variables and their values are the **UNIQUE** solution of the system of equalities obtained **AFTER** driving the non-basic variables to zero.

REMARK: NON-BASIC variables are ZERO by definition
but sometimes BASIC variables can be ZERO as well!

ADJACENCY

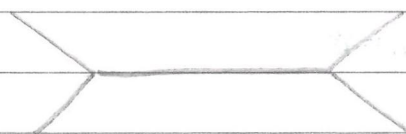
n dimensions

At each CPF set exactly n constraint boundaries meet
(each CPF/BFS is defined by exactly one basis)

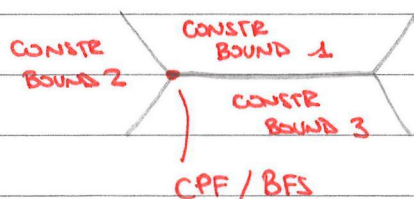
- Two CPF are ADJACENT if they have $n-1$ constraints boundaries in common

The two sets of non-basic variables defining the CPFs differ in exactly one variable

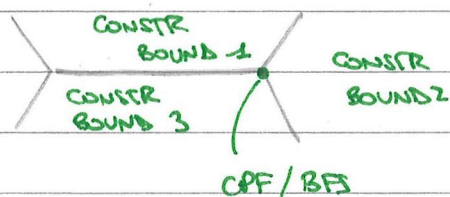
EXAMPLE = 3D



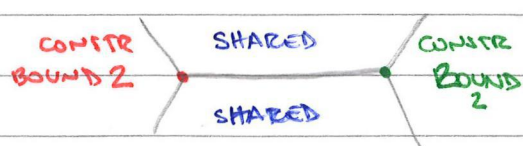
Horizontal line sticking further out of the page



the variables corresponding to
these 3 constraints are NON-BASIC



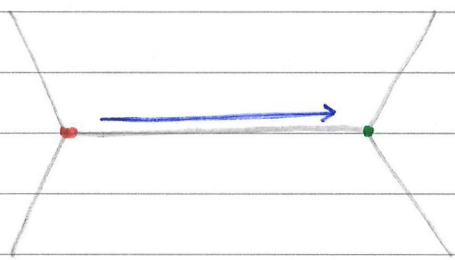
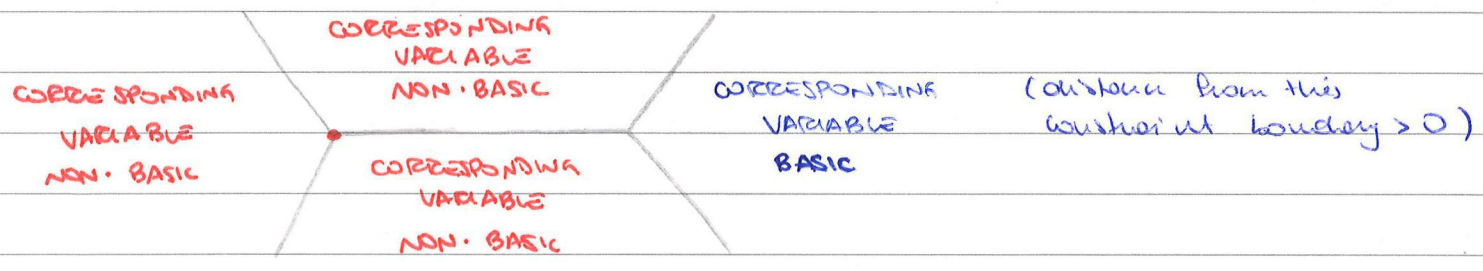
the variables corresponding to
these 3 constraints are NON-BASIC



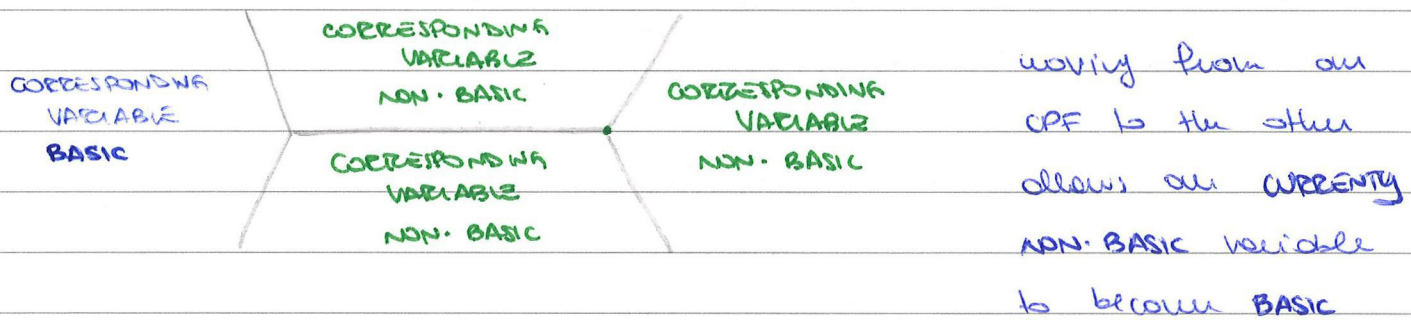
the GREEN CPF shares $n-1 = 2$
constraints with the RED CPF
 $\Rightarrow$ they are ADJACENT!

REMARK: $n-1$ common constraints means that the two adjacent CPF/BFF are connected by a line (EDGE), i.e., one dimensional structure

What changes from RED CPF to GREEN CPF?



move along the edge away from RED CPF until we reach the other constraint boundary (GREEN CPF)



(and vice versa our CURRENTLY BASIC variable to become NON-BASIC)

ENTERING variable: var. basic that becomes non basic (enters the BASIS)

LEAVING variable: basic that becomes non-basic (leaves the BASIS)

SIMPLEX METHOD

- Simplex method moves from CPF/BFS to an adjacent CPF/BFS
- Possibly it starts at the origin
(for the WINDR GLASS Co. example, it corresponds to $(0,0,4,12,12)$ with basis x_3, x_4, x_5)
- GREEDY: it moves to the neighboring CPF/BFS with the most potential to improve the objective function
(NEVER choose one that worsens the objective function)

1. How do you know when you've reached optimality?

- If the LP is feasible and has a bounded optimum, then exists at least one optimal. $\therefore$ CPF/BFS
- If none of the neighbors of the current CPF/BFS can improve the objective function, we know we have reached optimality!
 $\Rightarrow$ we just need to look at adjacent CPF/BFS!
 $\Rightarrow$ POWER of the SIMPLEX METHOD!

2. Can we reach an optimal CPF/BFS from ANY STARTING POINT?

YES!

There are no local optima 'per se'...

3. How do we choose the ENTERING basic variable?

PIVOTING RULES: we move in the direction with the most potential per unit increase to improve the objective function
GEOMETRICALLY: steepest angle of approach to the objective function

4. How do we choose the LEAVING variable?

Which constraint boundary we hit first?

$\Rightarrow$ the machinery of the SIMPLEX METHOD!

5. How do you ensure that the BASIS corresponds to a lin indep set of columns of the constraints matrix?

=> MACHINERY of the SIMPLEX METHOD

(the introduction of slack variables ensure full row rank)

